



SAPIENZA
UNIVERSITÀ DI ROMA

Project 11:

Uncertainty-Aware Road Obstacle Identification

Computer Vision Course 2024/2025

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Goals and task

Task

The focus is to develop a general framework for road obstacle identification, starting from semantic segmentation.

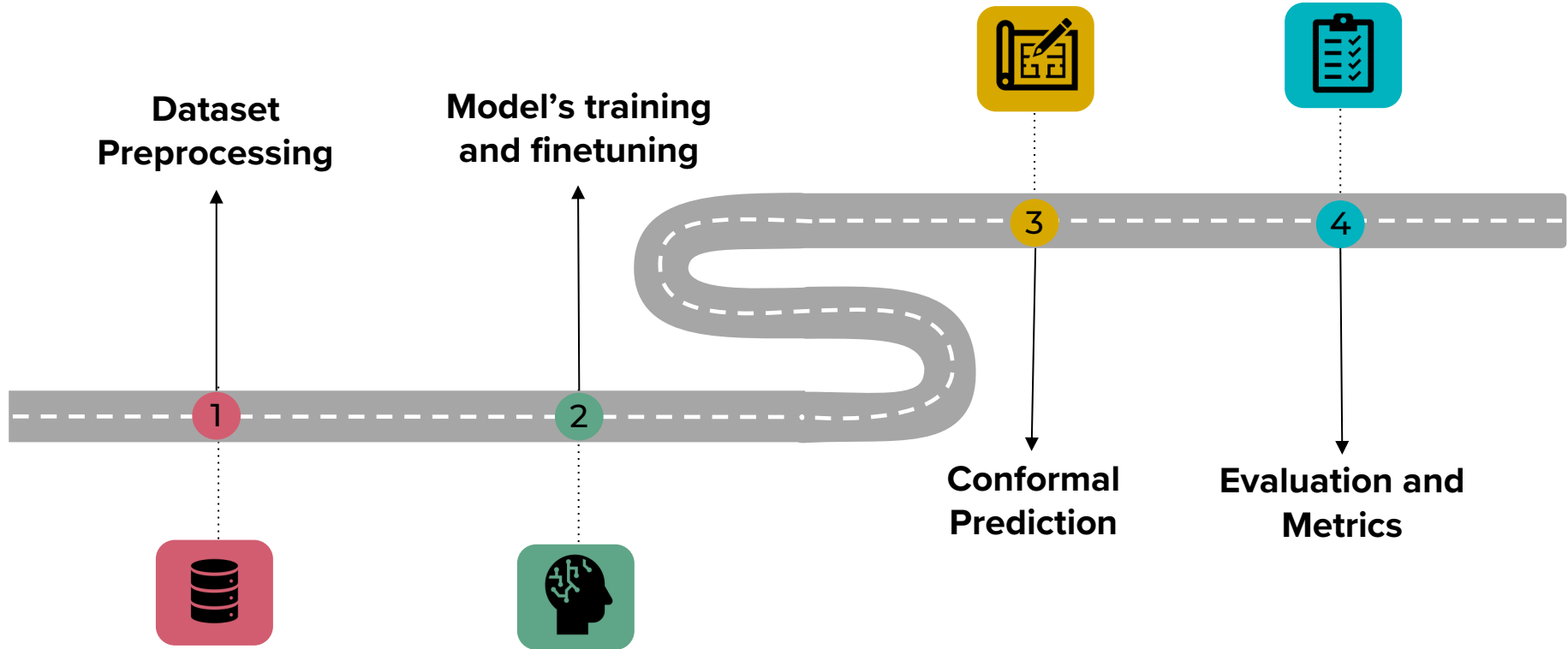
Allowing the model to also identify OOD objects.

Main Objective

- Anomaly aware Obstacle Segmentation
- Statistical Uncertainty Quantification
- Evaluation using detection metrics and uncertainty metrics



Project's workflow



Datasets used

Cityscapes

Composed of images and their segmentation masks. It was used for training.



1

RoadAnomaly

Composed of road images with OOD anomalies. It was used for evaluation.



3

LostAndFound

Composed of images with small objects in the street. It was used for finetuning.



2

FishyScapes

Composed of Cityscapes road images with OOD anomalies. It was used for evaluation.



4



Mapping of the Classes

The original Cityscapes classes were remapped into 8 macro-classes with a specific class dedicated to objects



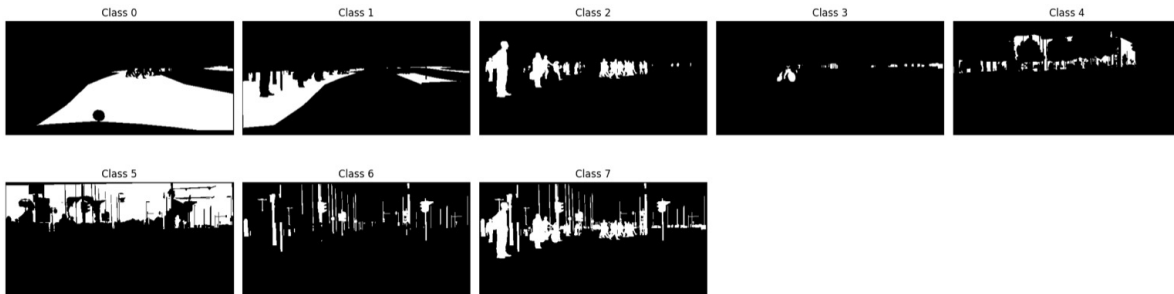
Transformations

With the Albumentation library all the images were preprocessed and normalized to be used as input for the model



One-Hot Mask

The label masks were converted into one-hot encoded tensors with 8 channels representing the 8 macro-classes



0=road, 1=pavement, 2=human, 3=vehicle, 4=building, 5=background, 6=traffic sign, 7=object



Transformations

With the Albumentation library all the images were preprocessed and normalized to be used as input for the model

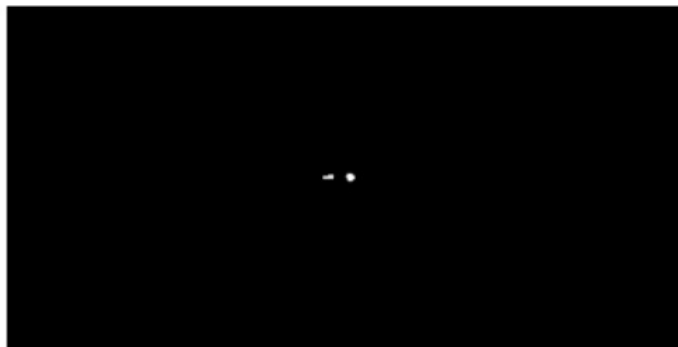


Transformed Image



One-Hot Mask

The label masks were converted into one-hot encoded tensors with 8 channels matching the Cityscapes macro-class structure



One-Hot mask

FishyScapes



This dataset was created by applying .png images of OOD anomalies to road images taken directly from the Cityscapes dataset. It was used for evaluation purposes only

RoadAnomaly



The images in this dataset were preprocessed and they were used for evaluation purposes only

The Model

Loss Function

Boundary Aware BCE loss with a lambda weight = 3.

Composed of:

- Standard BCE
- Boundary focused BCE

DeepLabv3 with ResNet50

Pretrained Model with ImageNet

Parameters

- Initial learning rate = 0.01
- Momentum = 0.9
- Weight decay = 0.0001
- Epochs = 15
- Boundary iteration = 5

Optimizer

Stochastic Gradient Descent with momentum and a poly decay learning rate

The Model

Training Phase

Batches of 4 images are given to the model which outputs 8 probability maps, one per each pixel.

Additionally, a boundary mask is computed to highlight object's outline.

The loss is computed using both the boundary mask and the full mask, assigning greater weight to boundary pixels.

Finetuning Phase

The trained model is initialized with a different loss function with a key change: it checks if at least one class is active in a pixel to check its validity.

The optimizer is reinitialized with a low learning rate.

The finetuning epochs are set to 5.



Model's Evaluation

Predicted Segmentation



Original Image - index 150



Predicted segmentation - image index 150



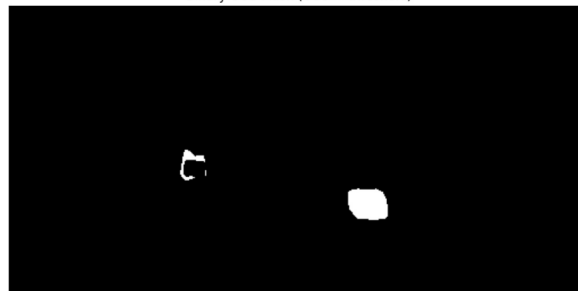
Unknown Object Mask



Original Image - index 150

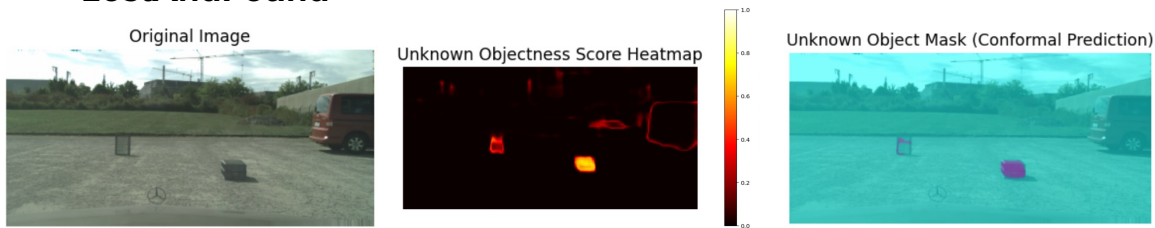


Binary UOS Mask (UOS > threshold)



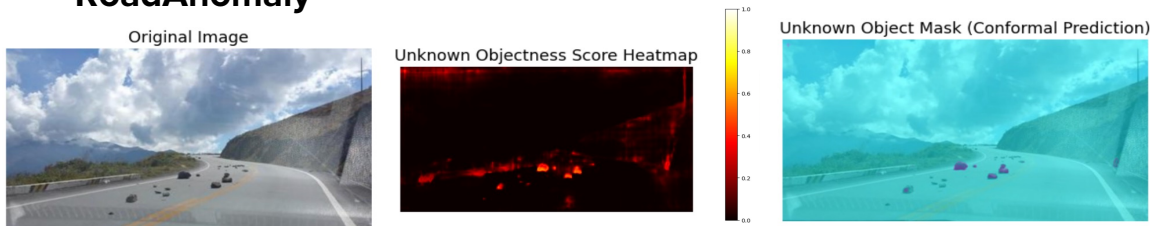
Conformal Prediction

- **LostAndFound**



During this phase a conformal prediction threshold was computed.

- **RoadAnomaly**



This allowed the detection of anomalous pixel with a statistical confidence of 99%.

- **Fishyscapes**



Any pixel with a UOS higher than the threshold is flagged as an anomaly

Results

Methods	LostAndFound Test		
With OOD data	FPR95	AP	AUROC
*Reference Paper	1.17	87.74	99.52
Our Method	41.97	16.68	92.29



Methods	CityScapes mIoU
*Reference Paper	76.85
Our Method	59.51

Mean mIoU	Pixel Accuracy	F1-Score
59.51	83.16	71.76

References

“

Chihiro Noguchi, Toshiaki Ohgushi, Masao Yamanaka. **Road Obstacle Detection based on Unknown Objectness Scores**

*Reference Paper

”

Luca Mossina, Joseba Daimau, Lèo Andrèol, **Conformal Semantic Image Segmentation: Post-hoc Quantification of Predictive Uncertainty**

“

Liang-Chieh,Chen, George Papandreou, Florian Schroff, Hartwig Adam, **Rethinking Atrous Convolution for Semantic Image Segmantion**

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Anastasios N. Angelopoulos and Stephen Bates, **A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification**



Thank you for your attention!

